

# Alejandro Morera Alvarez

AI LEAD AT PROFUTURO · ML FORESIGHT EXPERT AT NOVALAN

## EDUCATION

**PhD Researcher in Mathematics** January 2027

Technische Universität München

**Project:** *Learning and Dynamics in Stochastic Multi-Agent Systems* — joint project of the groups of Prof. Martin Bichler, Prof. Christoph Knochenhauer, Prof. Christian Kuehn, and Prof. Jingui Xie.

**Double M. Sc. in Mathematics and Applied Mathematics** April 2024 – August 2026

École Polytechnique Fédérale de Lausanne & Technische Universität München

German GPA 1.1 – “Passed with High Distinction”

Swiss GPA 5.65 – “Very Good”

**Highlights:** Stochastic Analysis, Fractal Geometry, (Random) Dynamical Systems, Optimal Transport.

**Thesis:** *Mean-Field Limits for Stochastic Interacting Particle Systems on Digraphs Measures: Convergence, Regularity and Applications to Machine Learning Dynamics on Large Networks*

**B. Sc. Mathematics** October 2022 – March 2024

Technische Universität München · German GPA 1.3

**Thesis:** *Borel Functional Calculus and the Spectral Theorem*

**B. Sc. Engineering Science** October 2018 – June 2023

Technische Universität München · German GPA 1.709

**Thesis:** *Borel Functional Calculus and the Spectral Theorem*

**American International School Salzburg** — AP Scholar with Distinction American GPA 4.031/4.3

**Colegio Vista Hermosa** — Duc in Altum Award (3 times) GPA 97.5/100

## WORK EXPERIENCE

**AI Lead** — Profuturo May 2026 – Now

Lead a team in strategic AI developments from the technical side and provide a bridge for communication between the team and several company areas, including legal, financial, risk, and investment.

**ML/AI Foresight Expert** — Novalan April 2026 – Now

Use ML/AI models and tools to predict desired phenomena; implement AI automations to build systems within the company which simplify analyses and processes.

**AI Analyst** — Profuturo August 2025 – May 2026

LLM, RAG and Automation Expert in the AI Team. Built RAG pipelines for AI agents with distinct tools, strategies, and architectures.

**Teaching Assistant** — EPFL September 2024 – June 2025

**Modules:** Probabilité (B. Sc. Mathématiques), Analysis III (B. Sc. Génie mécanique), Analyse IV (B. Sc. Physique).

**Teaching Assistant** — TUM October 2020 – August 2024

**Modules:** Probability Theory (M. Sc. Mathematics), Analysis 3 (B. Sc. Mathematics), Abstract Algebra (B. Sc. Mathematics), Analysis 1 (B. Sc. Physics), Real Analysis and Linear Algebra (B. Sc. Engineering Science).

RESEARCH EXPERIENCE

|                                                                                                                                                                             |                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| MASTER THESIS                                                                                                                                                               | February 2026 – August 2026    |
| Mean-Field Limits for Stochastic Interacting Particle Systems on Digraphs Measures: Convergence, Regularity and Applications to Machine Learning Dynamics on Large Networks |                                |
| PROJET DE MASTER                                                                                                                                                            | February 2025 – June 2025      |
| Fyodorov–Bouchaud Formula and Beyond                                                                                                                                        |                                |
| BACHELOR THESIS                                                                                                                                                             | April 2023 – June 2023         |
| Borel Functional Calculus and the Spectral Theorem for Bounded Operators                                                                                                    |                                |
| RESEARCH PROJECT                                                                                                                                                            | October 2023 – December 2023   |
| Convergence to Equilibrium: Entropy Methods in PDEs and Stochastic Analysis                                                                                                 |                                |
| R PROGRAMMING + RESEARCH PROJECT                                                                                                                                            | September 2023 – February 2024 |
| Education in Different Countries                                                                                                                                            |                                |

SEMINARS & WORKSHOPS

**Presented talks on** Elliptic Curves, Infinite-dimensional Vector Spaces, Selected Topics in Functional Analysis, Applied Asymptotic Analysis.

**Attended** Particle Systems in Random Environments (Summer School, 2024); research talks at EPFL (2024–2025) and ITAM (2025–2026).

LANGUAGES

|                  |              |             |                |                |
|------------------|--------------|-------------|----------------|----------------|
| Spanish – Native | English – C2 | German – C1 | French – B1-B2 | Norwegian – A2 |
|------------------|--------------|-------------|----------------|----------------|

COMPUTER & AI SKILLS

|             |                                                                                                                                                                                                                              |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GENERAL     | Autodesk AutoCAD Professional, Autodesk Inventor Professional (Computer-Aided Modelling of Products and Processes I), CATIA, Microsoft Office, MATLAB, Julia, and R.                                                         |
| AI & AGENTS | Claude Code- and AI-native development; Copilot Studio, Power Automate, n8n, Microsoft Azure AI, OpenAI AgentKit and Agents SDK, LangChain, LangGraph, LlamaIndex, tool/function calling, and MCP client/server integration. |
| HYBRID RAG  | Semantic vector search plus BM25/keyword retrieval; embeddings, document chunking, metadata filtering, query rewriting, reranking, context assembly, grounded citations, and RAG evaluation.                                 |
| DATA & APIS | SQL, PostgreSQL/pgvector, Qdrant, and Azure AI Search; FastAPI, Pydantic, REST/JSON APIs, OpenAI/Azure OpenAI APIs, Microsoft Graph/SharePoint integrations, OAuth 2.0, and API-key authentication.                          |
| ENGINEERING | Python, Pandas, document parsing/OCR, Git, Docker, automated testing, logging/observability, and secrets management.                                                                                                         |

OTHER EXPERIENCE

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Grupo ADO Summer Course                                                                                                                                                                                     | Mexico City, Mexico · June – July 2017 |
| Understanding of how the company works and how smaller subsections interact with each other to make the group as a whole work.                                                                              |                                        |
| Student Tutoring and Funding Administration                                                                                                                                                                 | May – July 2017                        |
| Administered funding for a student tutoring initiative, directly facilitating educational support for approximately 170 students across mathematics, physics, and chemistry, improving students’ pass rate. |                                        |
| Raised funds to build a well in a water-lacking community.                                                                                                                                                  |                                        |